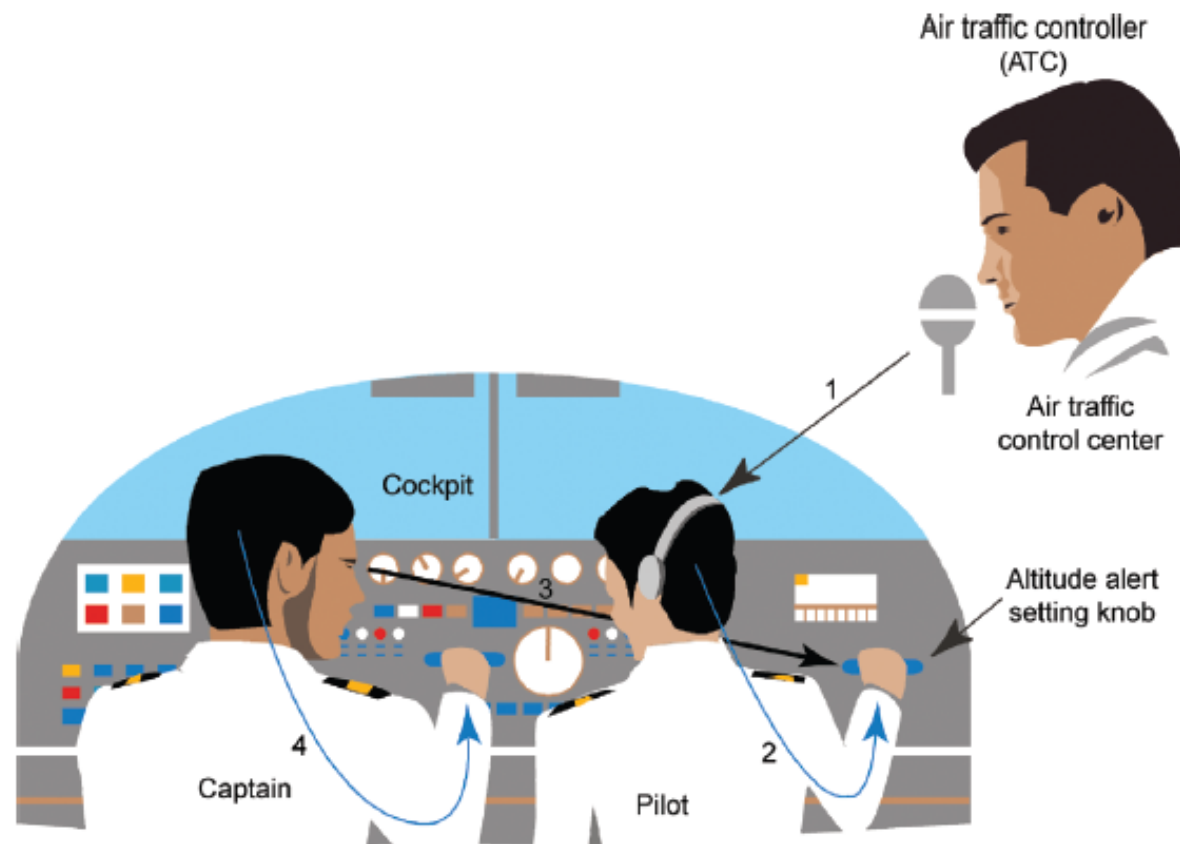


Distributed cognition

- Concerned with the nature of cognitive phenomena across individuals, artefacts, and internal and external representations (Hutchins, 1995)
- Describes these in terms of propagation across representational state
- Information is transformed through different media (computers, displays, paper, heads)



Propagation of representational states:

- 1 ATC gives clearance to pilot to fly to higher altitude (verbal)
- 2 Pilot changes altitude meter (mental and physical)
- 3 Captain observes pilot (visual)
- 4 Captain flies to higher altitude (mental and physical)

Figure 3.11 A cognitive system in which information is propagated through different media

Source: Preece, J. and Keller, L. (1994) *Human-Computer Interaction*, Figure 3.5 (p. 70) Addison Wesley, 1994.

How it differs from information processing

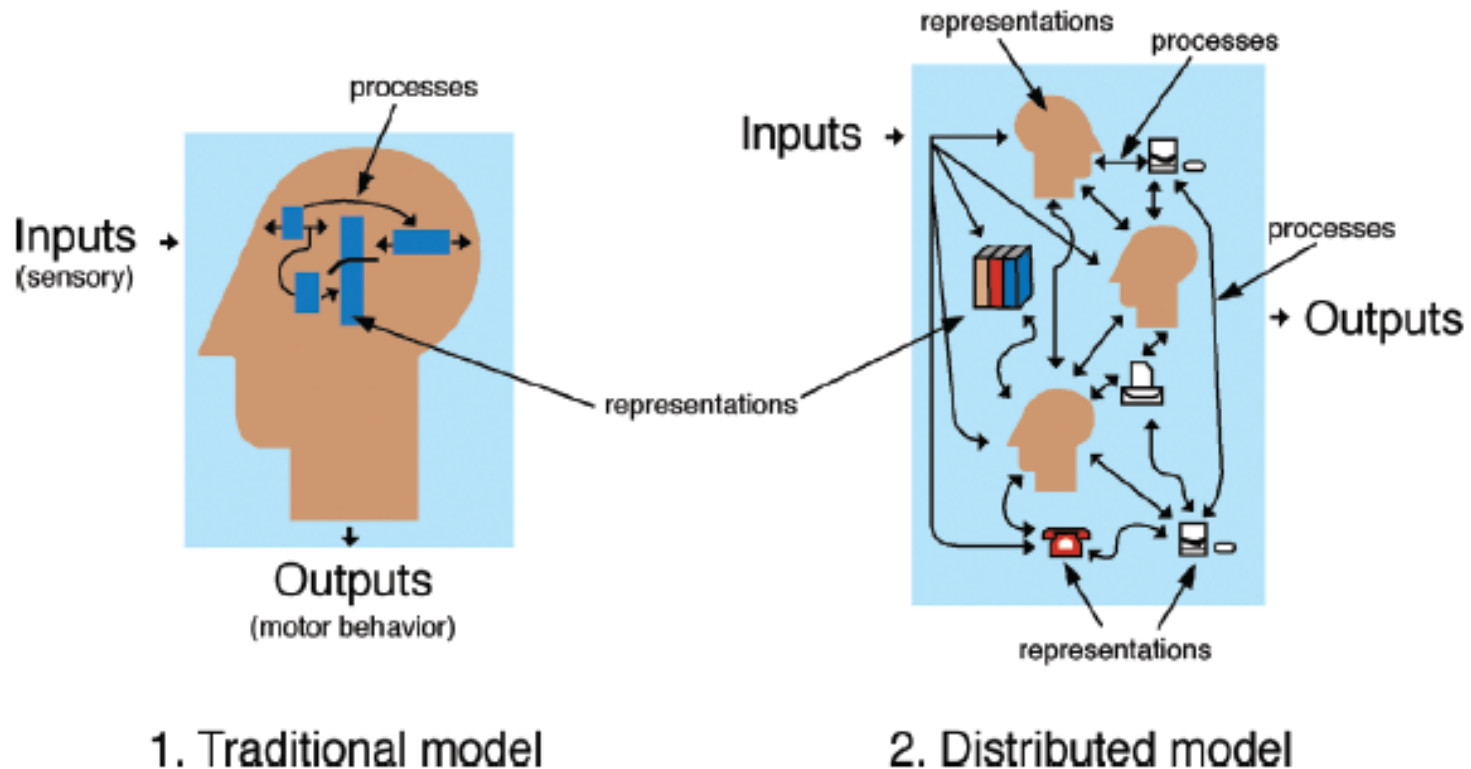


Figure 3.10 Comparison of traditional and distributed cognition approaches

What's involved

- The distributed problem-solving that takes place
- The role of verbal and non-verbal behavior
- The various coordinating mechanisms that are used (e.g. rules, procedures)
- The communication that takes place as the collaborative activity progresses
- How knowledge is shared and accessed